Pharmacy Distribution ERP — Project Discovery Report

Generated: 2026-10-15
Source of Truth: Verified source code inspection
Classification: Internal Technical Document

1. Identified Technology Stack

Backend

Component	Technology	Version
Language	Go	1.25.0
Web Framework	Gin	v1.12.0
ORM	GORM	v1.31.1
Database Driver	MySQL (`gorm.io/driver/mysql`)	v1.6.0
Authentication	JWT (`golang-jwt/jwt`)	v5.3.1
Caching	Redis (`redis/go-redis`)	v9.22.0
Logging	Zap (`uber/zap`)	v1.28.0
Validation	go-playground/validator	v10.30.3
Decimal	shopspring/decimal	v1.4.0
Module ID	`github.com/pixandco/erp-phrma`	—

Frontend

Component	Technology	Version
Framework	React	19.0.0
Build Tool	Vite	^8.0.16
Language	TypeScript	^7.0.1-rc
Styling	Tailwind CSS	^3.1.8
UI Primitives	Radix UI + shadcn	Latest
Data Fetching	TanStack Query	v5
Tables	TanStack Table	v8
Forms	React Hook Form + Zod	v7 + v3
HTTP Client	Axios	v1
State Management	Zustand	v5
Charts	ApexCharts, Recharts, Tremor	—
Animations	Framer Motion, dnd-kit	v12
Routing	React Router DOM	v6
Fonts	Inter, Geist Variable	—

Infrastructure

Component	Technology
Database	MySQL (dedicated DB per company)
Cache	Redis
Containerisation	Docker + Docker Compose
Frontend Web Server	Nginx

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2. Multi-Tenancy & Architecture

The system uses **database-per-company** multi-tenancy: - Platform database (`erp_platform`) stores company registrations - Each company gets its own MySQL database named from `company_code` - JWT claims embed `company_id`, `company_code`, `active_branch_id`, and `active_software_code` - `CompanyAuthMiddleware`

validates the JWT and resolves the company database connection on every request - All company tables also have `company_id` columns for belt-and-suspenders isolation

Request pipeline:

```
Browser → React SPA → Axios
  → Gin API (port 8080)
    → CORS → CompanyAuthMiddleware → BranchAccessMiddleware
    → SoftwareAccessMiddleware → RequirePermission
    → Handler → Service → Repository → MySQL (company DB) + Redis
    → Zap structured logging
```

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3. Module Implementation Status

Module	Backend	Frontend	Notes
Control Center	✅ IMPLEMENTED	✅ IMPLEMENTED	Full CRUD: company, branches, departments, designations, users, roles, permissions, access matrix, audit logs, login logs, approval workflows, document numbering, security settings
Finance	✅ IMPLEMENTED	✅ IMPLEMENTED	Financial years, CoA, journal entries, payment/receipt vouchers, bank accounts, cheque books, bank transactions, reconciliations, petty cash, fixed assets, general ledger, reports
Inventory	✅ IMPLEMENTED	✅ IMPLEMENTED	Warehouses, product master, products, batches, stock balances/ledger, opening stock, GRNs, transfers, adjustments, purchase/sales returns
Invoice Center	✅ IMPLEMENTED	✅ IMPLEMENTED	Customers, sales orders, sales invoices, credit notes, debit notes, customer receipts, finance settings, posting integration, print formats, reports
Compliance Center	❌ NOT IMPLEMENTED	✅ FRONTEND ONLY	Licenses, batch recall, batch hold, expiry disposal, regulatory — pages exist but no backend API
Platform Admin	✅ IMPLEMENTED	N/A	Company provisioning

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4. Database Tables Summary (80+ tables verified)

Control Center / Company Core

`companies`, `branches`, `departments`, `designations`, `designation_default_roles`, `designation_departments`, `users`, `roles`, `permissions`, `role_permissions`, `user_branch_access`,

`user_branch_roles`, `user_software_access`, `user_organization_assignments`, `software_modules`, `audit_logs`, `login_logs`, `password_reset_tokens`, `system_settings`, `approval_workflows`, `approval_workflow_versions`, `approval_workflow_stages`, `document_numbering_rules`, `document_number_sequences`, `security_policies`, `trusted_ip_rules`, `backup_policies`, `notification_rules`

Finance (37 tables)

`financial_years`, `accounting_periods`, `account_classifications`, `account_groups`, `chart_of_accounts`, `opening_balances`, `tax_settings`, `journal_entries`, `journal_entry_lines`, `journal_entry_approvals`, `journal_entry_reversals`, `payment_vouchers`, `payment_voucher_lines`, `payment_voucher_approvals`, `receipt_vouchers`, `receipt_voucher_lines`, `receipt_voucher_approvals`, `bank_accounts`, `cash_accounts`, `cheque_books`, `cheque_leaves`, `bank_transactions`, `bank_reconciliations`, `bank_reconciliation_lines`, `petty_cash_funds`, `petty_cash_vouchers`, `petty_cash_voucher_lines`, `petty_cash_voucher_approvals`, `petty_cash_replenishments`, `petty_cash_replenishment_approvals`, `fixed_asset_categories`, `fixed_assets`, `fixed_asset_depreciation_runs`, `fixed_asset_depreciation_lines`, `fixed_asset_disposals`, `fixed_asset_disposal_approvals`, `general_ledger_entries`

Inventory (31 tables)

`warehouses`, `warehouse_locations`, `product_categories`, `product_units`, `dosage_forms`, `generic_names`, `manufacturers`, `suppliers`, `products`, `product_barcodes`, `product_batches`, `stock_balances`, `stock_ledger_entries`, `opening_stock_entries`, `opening_stock_entry_lines`, `opening_stock_entry_approvals`, `goods_receipt_notes`, `goods_receipt_note_lines`, `goods_receipt_note_approvals`, `stock_transfers`, `stock_transfer_lines`, `stock_transfer_approvals`, `stock_adjustments`, `stock_adjustment_lines`, `stock_adjustment_approvals`, `purchase_returns`, `purchase_return_lines`, `purchase_return_approvals`, `sales_returns`, `sales_return_lines`, `sales_return_approvals`

Invoice Center (22 tables)

`customer_categories`, `customers`, `customer_addresses`, `customer_contacts`, `sales_orders`, `sales_order_lines`, `sales_order_approvals`, `sales_invoices`, `sales_invoice_lines`, `sales_invoice_approvals`, `credit_notes`, `credit_note_lines`, `credit_note_approvals`, `debit_notes`, `debit_note_lines`, `debit_note_approvals`, `customer_receipts`, `customer_receipt_lines`, `customer_receipt_approvals`, `invoice_center_finance_settings`, `invoice_center_finance_postings`, `invoice_print_formats`

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5. Document Lifecycle Pattern

All transactional documents follow a consistent two-track lifecycle:

Approval Track: `draft → submitted → approved / rejected`
Posting Track: `unposted → posted`

Posting requires prior approval and irreversibly commits changes to the general ledger and/or stock ledger.

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6. Permission System

Permissions follow the pattern `module.resource.action` (e.g., `finance.journal.approve`, `inventory.grn.post`).

Middleware checks permissions on every protected route: - `RequirePermission("finance.journal.approve")` — checks user's role-based permissions - Branch access is also enforced via `BranchAccessMiddleware` - Software module access is enforced via `SoftwareAccessMiddleware`

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7. Known Gaps & Technical Risks

Item	Severity	Details
Compliance Center backend missing	HIGH	6 frontend routes have no API
No Purchase Order entity	MEDIUM	GRNs reference PO by free text only
Customer balance denormalisation	MEDIUM	`customers.current_balance` must stay in sync with postings
GL running balance	MEDIUM	Denormalised field; rebuild endpoint exists
No supplier bill/AP table	LOW	Payments handled through payment vouchers
2FA model exists, logic unverified	LOW	`two_factor_enabled` flag in user model
No inventory valuation report	LOW	No API endpoint found
No VAT/SSCL tax-specific report	LOW	No tax report endpoint found